

AI Application & Letter Assistant

Complete Project Blueprint & Development Roadmap

Purpose	A strong full-stack React project blueprint to build after learning TypeScript and Supabase.
Target Stack	React + TypeScript + Supabase + PostgreSQL + AI + PDF Generation
Project Goal	Solve a real-world problem by helping users create professional formal applications and letters through guided workflows.

Prepared as a personal reference copy for future project development.

1. Project Vision

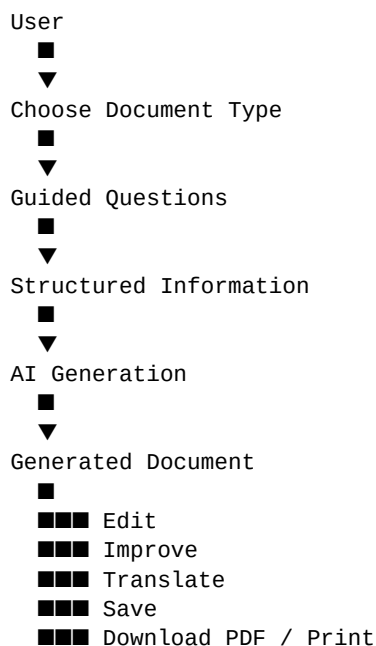
Project Name: AI Application & Letter Assistant

Core Idea: Help people who know what they want to say but do not know how to write a proper formal application or letter.

The most important product principle is simple:

The user should never need to understand prompt engineering. Instead of asking them to write a complicated AI prompt, the application collects structured information through guided forms and transforms it into a professional document.

Core Product Flow



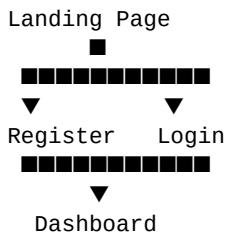
2. Authentication

Authentication is part of the real product rather than just a learning exercise.

V1 Features

- User Registration
- Login
- Logout
- Persistent Session
- Protected Dashboard
- Basic User Profile

Authentication Flow



3. Landing Page

The landing page should immediately explain what the product does. The design should feel modern, trustworthy, and approachable rather than overloaded with AI branding.

AI WriteAssist

Write any formal application
without the stress.

Tell us what you need.
We'll help you create a clear,
professional document.

[Create Application]

- ✓ Guided Forms
- ✓ AI Assisted Writing
- ✓ Multiple Languages
- ✓ Download PDF

4. Dashboard

The dashboard becomes the user's workspace where they can create new applications and manage previously saved documents.

Dashboard

Hello, User ■

[+ Create New]

Recent Applications

- Leave Application
Education
Open / Edit / Delete
- Mobile Number Change
Banking
Open / Edit / Delete

5. Application Categories

Do not create hundreds of categories. A carefully chosen set of broad categories is enough for Version 1.

Category	Examples
Education	Leave application, Fee concession, Bonafide request

Workplace	Leave request, Resignation, Work from home
Banking	Mobile change, Address change, Name correction
Government / Official	Certificate request, General application
Housing / Society	Maintenance request, Society permission
Healthcare	Medical leave, Certificate request
Complaint / Request	General complaint or formal request
General Letter	Any formal letter outside predefined categories

6. Guided Form — The Heart of the Project

This is the feature that differentiates the project from a normal AI chatbot.

Instead of showing one empty prompt box, the user completes a structured multi-step form.

Example: Banking → Mobile Number Change

Step 1 / 3

Your Name
Bank Name
Branch
Account Number

Continue

Step 2 / 3

Old Mobile Number
New Mobile Number
Reason for Change

Continue

Step 3 / 3

Recipient: Branch Manager
Language: English

[Generate Application]

This workflow dramatically reduces user confusion and creates more reliable AI output.

7. AI Document Generation

AI should be used carefully and only where it adds real value.

Input

Document Type
+

User Information
+
Application Details
+
Language
+
Tone

Output

Professional Formal Application

To,
The Branch Manager
...

Subject: Request for Mobile Number Change

Respected Sir/Madam,

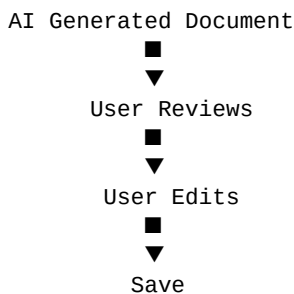
[Professionally generated body]

Yours faithfully,
User

The AI should generate polished content from structured information rather than inventing unnecessary facts.

8. Edit Generated Document

Never treat AI output as the final answer. The user must remain in control.



Improve My Draft

A user may write a rough sentence such as:

"Sir my mobile number is not working please change my number in account."

The Improve Draft feature rewrites it into a formal professional application while preserving the original meaning.

Tone Options

- Simple
- Formal
- Very Formal
- Polite

9. Language Support

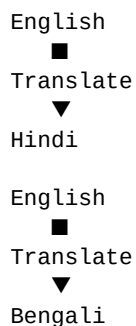
Keep Version 1 focused.

- English
- Hindi
- Bengali

The architecture should allow additional languages later without redesigning the application.

10. Translation

Users should be able to translate an existing saved application when required.



Translation should happen only on demand so AI usage remains controlled.

11. Application History

Every authenticated user owns their own saved applications.

History Features

- View all applications
- Filter by category
- Open existing document
- Edit saved application
- Delete application
- Sort by recent activity

My Applications

All | Education | Banking | Workplace

Leave Application

27 Aug 2026

[Open] [Edit] [Delete]

- Print
- Copy text

Formal Application Preview

To,
The Branch Manager

Subject: Request for ...

Respected Sir/Madam,

.....

Date:

Signature:

15. Optional Future Features

These are valuable ideas, but they belong to Version 2 rather than the initial build.

Feature	Purpose
Upload Existing Application	Improve previously written documents
OCR	Extract text from image or scanned PDF
Voice Input	Speak the request instead of typing
Document Type Detection	AI suggests the correct application type
More Languages	Expand regional accessibility
Advanced Templates	Dynamic institution-specific workflows

16. Features to Avoid in Version 1

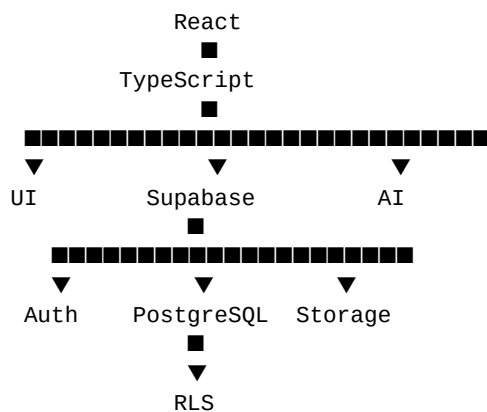
Keeping the project focused is one of the most important engineering decisions.

- Citizen eligibility engine
- Government scheme database
- Real-time government information
- Legal advice
- Document verification
- Complex OCR pipeline
- Voice assistant

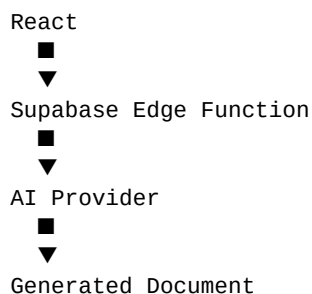
- General AI chatbot
- Payment system
- Admin marketplace
- 50+ languages
- 100+ document templates

The project becomes stronger when a few features are polished exceptionally well instead of many features being incomplete.

17. Technical Architecture



AI Flow



The important security principle is that the AI secret key should never live directly inside the React frontend.

18. TypeScript's Role

TypeScript should support the project rather than make it more complicated.

Example Application Type

```

type Application = {
  id: string;
  userId: string;
  title: string;
  category: string;
  documentType: string;
  content: string;
  language: string;
  createdAt: string;
}
  
```

```
updatedAt: string;
```

Example User Type

```
type UserProfile = {
  id: string;
  name: string;
  email: string;
  phone?: string;
};
```

These simple interfaces are enough for the beginning. Advanced TypeScript concepts can be introduced only when the project naturally requires them.

19. Responsive Design

The application should work beautifully across devices.

- Desktop
- Laptop
- Tablet
- Mobile

A user should comfortably create and download an application from a phone as well as from a desktop browser.

20. UI / UX Direction

The product should feel professional because users may create documents for banks, schools, workplaces, housing societies, and government offices.

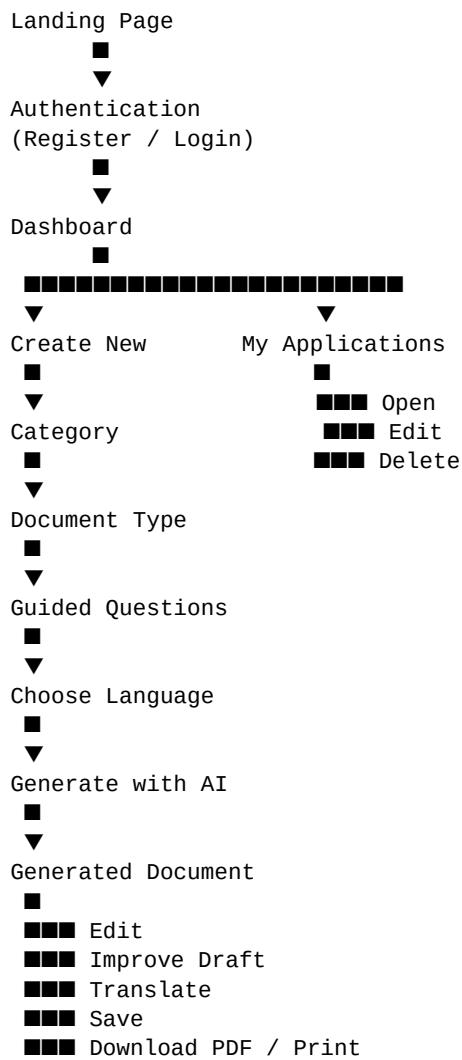
Design Principles

- Clean visual hierarchy
- Minimal clutter
- Trustworthy typography
- Guided user journey
- Accessible responsive layout
- Professional document preview

What do you need?

Sector	Percentage
Banking	10%
Education	10%
workplace	10%

21. Complete Version 1 Blueprint



22. Version 1 vs Version 2

Feature	Version
Authentication	V1
Supabase Auth	V1
User Profile	V1
Guided Forms	V1
AI Generation	V1
Edit Document	V1
Application CRUD	V1

Row Level Security	V1
English / Hindi / Bengali	V1
PDF Download	V1
Copy / Print	V1
Improve Draft	V1
Voice Input	V2
OCR	V2
Upload Existing Document	V2
AI Document Type Detection	V2
Additional Languages	V2
Advanced Dynamic Templates	V2
CitizenAssist Module	Not Initially

23. Recruiter Story

A polished explanation of the project:

I built a full-stack AI-powered document assistant that helps users create formal applications and letters through guided workflows instead of requiring prompt engineering knowledge.

Technical Talking Points

- React frontend
- TypeScript for type-safe development
- Supabase Authentication
- PostgreSQL database
- Row Level Security for user data isolation
- CRUD application management
- AI-powered document generation and improvement
- Multilingual document support
- PDF generation for real-world usability
- Responsive user interface

24. Learning & Development Roadmap

Do not start building the complete project immediately. Build your skills in the correct order.

NOW

1. Finish TypeScript Fundamentals



2. Learn TypeScript with React



3. Learn Supabase

- Authentication
- PostgreSQL
- CRUD
- Row Level Security



4. Start AI Application & Letter Assistant

- Database Design
- Folder Structure
- UI Design
- Authentication
- Guided Forms
- CRUD
- AI Integration
- PDF Generation
- Deployment

25. Final Project Discipline

The goal is not to keep searching for new project ideas. The goal is to turn one strong idea into a polished portfolio-quality product.

Build Version 1 completely. Test it carefully. Improve the design. Deploy it. Only then consider Version 2 features such as OCR, voice input, or intelligent document detection.

End of Blueprint — AI Application & Letter Assistant